

ARDUINO BASED CNC MACHINE

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Abstract - Due to the rapid growth of technology the usage & utilization of CNC machine in industries are increased. The fabrication of low cost CNC machine is used to reduce cost and complexity of machine. This paper deals with the design of automatic mini CNC machine for machining wood, plastic, foam and soft metals. The Idea behind our project is to design and drill PCB based on low cost CNC system the lower cost is achieved by incorporating features of PC with ARDUINO UNO controller in an arduino. Key Words: CNC, Arduino controller, G code,

I. INTRODUCTION

The CNC machine comprises of the mini computer or the microcomputer that acts as the controller unit of the machine. CNC Machining is a process used in the manufacturing sector that involves the use of computers to control machine tools. A computer program is customized for an object and the machines are programmed with CNC machining language (called G-code) that essentially controls all features like feed rate, coordination, location and speeds. With CNC machining, the computer can control exact positioning and velocity. CNC machining is used in manufacturing both metal and plastic parts. The work includes design & development of prototype modelling of CNC machine tool controller unit. In modern CNC systems, end-to-end component design is highly automated using computer- programs produce a computer file that is interpreted to Extract the commands needed to operate an aided design (CAD) and computer-aided manufacturing (CAM) programs. In this paper, a low-cost, desktop prototype 3- axis vertical CNC mill is developed for purposes of student experiments in CAD/CAM and CNC programming areas. Open source microcontroller platform Arduino is used for control of the motors, and open source software is used for executing the G code and M code for machining applications. The Arduino platform has become quite popular with people just starting out with electronics, and for good reason. Unlike most previous programmable circuit boards, the Arduino does not need a separate piece of hardware (called a programmer) in order to load new code onto the board you can simply use a USB cable. Under CNC Machining, machine tools function through numerical control.

II. OBJECTIVE

The main objective is to make a low cost CNC machine which can machine wood, plastic, foam and soft metals. Reduces cost of machine and increases the flexibility. To work in more than 2axis.

III. PROBLEM DEFINITION

The available Arduino controlled CNC machines are having only 2 axis movement. The structure is weak and can machine foam only.

IV. BLOCK DIAGRAM

Fig. 1 Magnetization as a block diagram of CNC machine.

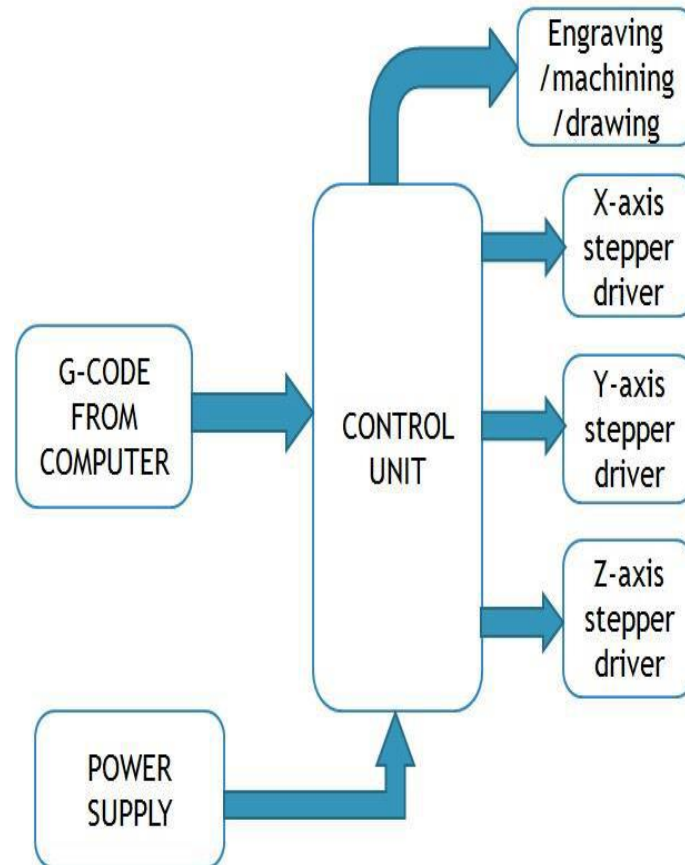


Fig-1 block diagram

1.1 G CODE FROM PC:

G-Code is one of a number of computer code languages that are used to instruct CNC machining devices what motions they need to perform such as work coordinates, canned cycles, and multiple repetitive cycles.

1.2 CONTROL UNIT:

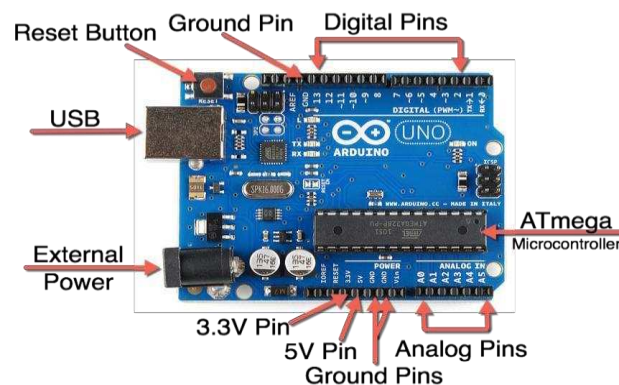


Fig-2 Arduino UNO

This system uses Arduino controller platform with Arduino UNO. This board is the brain of our CNC machine and serves as the main interface to the Computer. The Arduino board is programmed with a modified Riprap G-Code interpreter and the 3 axes motion.

1.3 STEPPER MOTOR (OLD CD ROM):

A stepper motor is a brushless, synchronous electric motor that converts digital pulses into mechanical shaft rotation in a number of equal steps. Used to control the movement of work table in X, Y & Z axes.

1.4 EASY DRIVER BOARD:

The Easy Driver is a simple to use stepper motor driver, compatible with anything that can output a digital 0 to 5V pulse.

1.5 DC MOTOR:

It helps to run the spindle of the machine which in turn rotates the cutting tool. The spindle consists of a DC motor. The main function of it is to cut metal with high speed and accuracy. Motors are available with suitable adaptors for different requirements.



Fig-3 DC motor

III. PROCESS DESCRIPTION

Main blocks of this system consist of power supply, 3 Easy drivers each connected to stepper motors X, Y, Z. From power supply we get two voltages i.e. +5V and +12V. 5V which is required to Arduino UNO, Easy drivers whereas +12V supply is required to stepper motors. The job is fed through the computer software to the Arduino. With this data the stepper motors of the CD Rom start by the driver circuitry and horizontal and vertical movement of work table is obtained. The spindle fastened to the vertical axis can remove material with proper feed and depth of cut.

IV. DESIGN OF CNC MACHINE

There are 3 movements of using 3 CD ROMs. The horizontal movement (X) i.e. forward & backward movement is provided by the lower CD Rom. The 2nd CD Rom is mounted between the 2 columns which provides side movements (Y) i.e. left and right hand side movements. The spindle which is mounted on the 3rd CD Rom provides vertical movement (Z) for feed of tool.

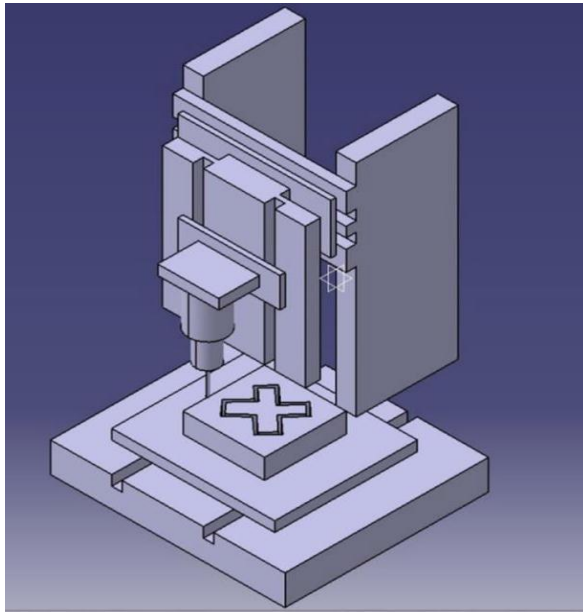


Fig-4.1 front view of cnc machine design

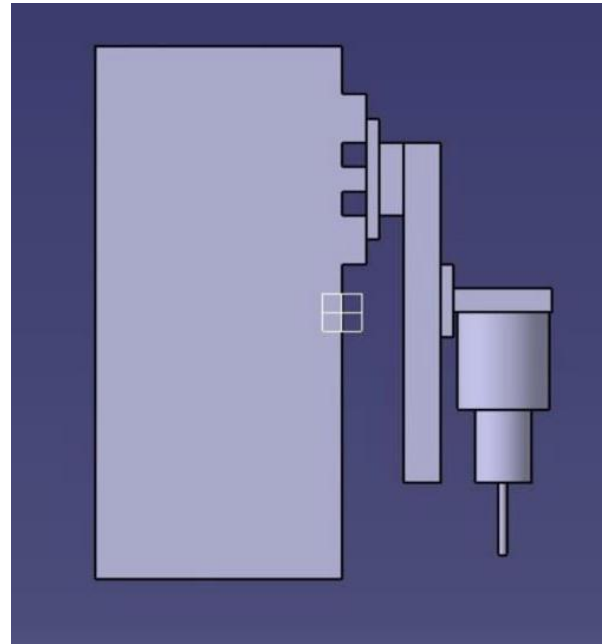


Fig-4.2 side view of cnc machine design

V. CONCLUSION

This setup of hardware with a combination of G-code gives better accuracy and reduces the work load. G code make easy to find the information of locations of all stepper motor moving, as the status of our moving motor are directly seen on computer hence we can start or stop the machine whenever we are needed. Making a small machine brings a flexibility to do work.

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